

ELCT 361
Solutions for HW #1

①

Q1.

(a) "Electromagnetics" is the study of the effects of charges at rest and in the motion.

(b) M (Meter), [m], Length
K (Kilogram) [kg], Mass
S (Second) [Sec.], Time
A (Ampere) [A], Current

(c) $\vec{E}$ (Electric field), [V/m]

$\vec{D}$ (Electric flux Density), or (Electric Displacement), [C/m²]

$\vec{H}$ (Magnetic field Intensity), [A/m]

$\vec{B}$ (Magnetic flux Density), [T] or [Wb/m²]

(d)

(i) Velocity of light in free space, c [m/sec], $c = 3 \times 10^8$

(ii) permeability of free space, $\mu_0 = 4\pi \times 10^{-7}$ [H/m]

(iii) permittivity of free space, $\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} = 8.854 \times 10^{-12}$ [F/m]

$$\begin{aligned} \epsilon_0 ? \quad \boxed{c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}} &\rightarrow \epsilon_0 = \frac{1}{c^2 \mu_0} = \frac{1}{(3 \times 10^8)^2 \times (4\pi \times 10^{-7})} \\ &= \frac{1}{36\pi} \times 10^{-9} \\ &\approx 8.854 \times 10^{-12} \text{ [F/m]} \end{aligned}$$

Q2.

$$M(-1, 2, 1), N(3, -3, 0), P(-2, -3, -4)$$

$$\vec{AM} = \vec{AOM} = -\hat{ox} + 2\hat{oy} + \hat{oz}, \quad \vec{AN} = \vec{AON} = 3\hat{ox} - 3\hat{oy}, \quad \vec{AP} = \vec{AOP} = -2\hat{ox} - 3\hat{oy} - 4\hat{oz}$$

$$(a) \vec{AMN} = \vec{AN} - \vec{AM} = 4\hat{ox} - 5\hat{oy} - \hat{oz}$$

$$|\vec{AMN}| = \sqrt{(4)^2 + (-5)^2 + (-1)^2} = \sqrt{42}$$

$$\hat{AMN} = \frac{\vec{AMN}}{|\vec{AMN}|} = \frac{1}{\sqrt{42}} (4\hat{ox} - 5\hat{oy} - \hat{oz})$$

$$\begin{aligned} (b) \vec{AMN} + \vec{AMP} &= (\vec{AN} - \vec{AM}) + (\vec{AP} - \vec{AM}) \\ &= (4\hat{ox} - 5\hat{oy} - \hat{oz}) + (-\hat{ox} - 5\hat{oy} - 5\hat{oz}) \\ &= 3\hat{ox} - 10\hat{oy} - 6\hat{oz} = \underline{3\hat{ox} - 10\hat{oy} - 6\hat{oz}} \end{aligned}$$

$$(c) |\vec{AM}| = \sqrt{(-1)^2 + (2)^2 + (1)^2} = \sqrt{6}$$

$$|\vec{AN}| = \sqrt{(3)^2 + (-3)^2} = \underline{3\sqrt{2}}$$

$$\begin{aligned} (d) |5\vec{AP} - 2\vec{AN}|; (5\vec{AP} - 2\vec{AN}) &= (-10\hat{ox} - 15\hat{oy} - 20\hat{oz}) - (6\hat{ox} - 6\hat{oy}) \\ &= \underline{-16\hat{ox} - 9\hat{oy} - 20\hat{oz}} \end{aligned}$$

Hence,

$$|5\vec{AP} - 2\vec{AN}| = \sqrt{(-16)^2 + (-9)^2 + (-20)^2} \approx \underline{18.47}$$

$$\begin{aligned} (\vec{AMP} + \vec{APN}) &= (\vec{AP} - \vec{AM}) + (\vec{AN} - \vec{AP}) = (-\hat{ox} - 5\hat{oy} - 5\hat{oz}) + (5\hat{ox}) \\ &= \underline{4\hat{ox} - 5\hat{oy} - 5\hat{oz}} \end{aligned}$$

(3)

Therefore,

$$\begin{aligned} |5\vec{A}_p - 2\vec{A}_N|(\vec{A}_{MP} + \vec{A}_{PN}) &= (18.47)(4\hat{a}_x - 5\hat{a}_y - 5\hat{a}_z) \\ &= \underline{\underline{73.88\hat{a}_x - 92.35\hat{a}_y - 92.35\hat{a}_z}} \end{aligned}$$

Q3.

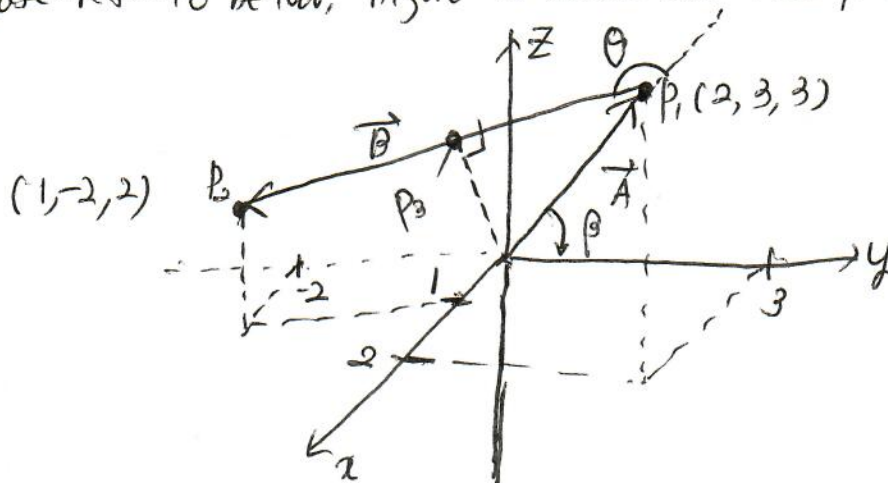
(a) Vector $\vec{A}$ is given by the distance vector from $O(0,0,0)$ to $P_1(2,3,3)$. Thus,

$$\begin{aligned} \vec{A} &= \underline{\underline{2\hat{a}_x + 3\hat{a}_y + 3\hat{a}_z}} \\ |\vec{A}| &= \sqrt{(2)^2 + (3)^2 + (3)^2} = \underline{\underline{\sqrt{22}}} \\ \hat{a}_A &= \frac{\vec{A}}{|\vec{A}|} = \underline{\underline{\frac{1}{\sqrt{22}}(2\hat{a}_x + 3\hat{a}_y + 3\hat{a}_z)}} \end{aligned}$$

(b) The angle β between $\vec{A}$ and the y -axis is obtained from

$$\begin{aligned} \vec{A} \cdot \hat{a}_y &= |\vec{A}| \cos \beta \\ \beta &= \cos^{-1} \left(\frac{\vec{A} \cdot \hat{a}_y}{|\vec{A}|} \right) = \cos^{-1} \left(\frac{3}{\sqrt{22}} \right) = \underline{\underline{50.2^\circ}} \end{aligned}$$

please refer to below figure to understand this problem.



(C) As can be seen & understood from the Figure,

$$\vec{B} = (1-2)\hat{a}_x + (-2-3)\hat{a}_y + (2-3)\hat{a}_z = \underline{\underline{-\hat{a}_x - 5\hat{a}_y - \hat{a}_z}}$$

$$(d) \theta = \cos^{-1} \left[\frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} \right] = \cos^{-1} \left[\frac{(-2-15-3)}{\sqrt{22} \cdot \sqrt{27}} \right] \approx \underline{\underline{145.1^\circ}}$$

(e) Refer to the figure in page ②, the perpendicular distance between the origin and vector $\vec{B}$ is the distance $|\vec{OP}_3|$.

From right triangle, OP_3

$$|\vec{OP}_3| = |\vec{A}| \sin(180^\circ - \theta) = \sqrt{22} \sin(180 - 145.1) = \underline{\underline{2.68}}$$

$$(f) \vec{A} = 2\hat{a}_x + 3\hat{a}_y + 3\hat{a}_z, \quad \vec{B} = (-\hat{a}_x - 5\hat{a}_y - \hat{a}_z)$$

$$\vec{A} \cdot \vec{B} = (-2-15-3) = \underline{\underline{-20}} \quad \left. \begin{array}{l} \vec{B} \cdot \vec{A} = (-2-15-3) = -20 \end{array} \right\} \text{Exactly the same indicating that}$$

"commutative law" is valid.

$$(g) \vec{A} \times \vec{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 2 & 3 & 3 \\ -1 & -5 & -1 \end{vmatrix} = (-3+15)\hat{a}_x + (-5+2)\hat{a}_y + (-10+3)\hat{a}_z \\ = \underline{\underline{12\hat{a}_x - 3\hat{a}_y - 7\hat{a}_z}}$$

$$\vec{B} \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ -1 & -5 & -1 \\ 2 & 3 & 3 \end{vmatrix} = (-15+3)\hat{a}_x + (-2+3)\hat{a}_y + (-3+10)\hat{a}_z \\ = \underline{\underline{-12\hat{a}_x + \hat{a}_y + 7\hat{a}_z}}$$

$\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$, Not Valid for "commutative Law"

(5)

Q4.

$$\vec{A} = \hat{a}_x - \hat{a}_y + 2\hat{a}_z$$

$$\vec{B} = \hat{a}_y + \hat{a}_z$$

$$\vec{C} = -2\hat{a}_x + 3\hat{a}_z$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -3\hat{a}_x - \hat{a}_y + \hat{a}_z$$

$$(\vec{A} \times \vec{B}) \times \vec{C} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ -3 & -1 & 1 \\ -2 & 0 & 3 \end{vmatrix} = \underline{\underline{-3\hat{a}_x + 7\hat{a}_y - 2\hat{a}_z}}$$

Similar way, $\vec{A} \times (\vec{B} \times \vec{C}) = \underline{\underline{2\hat{a}_x + 4\hat{a}_y + \hat{a}_z}}$

therefore,

$$(\vec{A} \times \vec{B}) \times \vec{C} \neq \vec{A} \times (\vec{B} \times \vec{C}),$$

The fact that the results of two vector triple products are different is a demonstration of the inequality stated in our lecture presentation material (chapter, 2-1)

Q5.

Let, $\vec{B} = B_x \hat{a}_x + B_y \hat{a}_y + B_z \hat{a}_z$. We know that

$$|\vec{B}| = (B_x^2 + B_y^2 + B_z^2)^{1/2} \quad (\text{unit vector})$$

(6)

(a) $\vec{B} \parallel \vec{A}$ requires $\vec{B} \times \vec{A} = 0$

$$\vec{A} \times \vec{B} \text{ or } \vec{B} \times \vec{A} = 0$$

$$-2B_z - B_y = 0$$

$$B_x - 5B_z = 0$$

$$5B_y + 2B_x = 0$$

combining with $B_x^2 + B_y^2 + B_z^2 = 1$

$$B_x = \frac{5}{\sqrt{30}}, \quad B_y = -\frac{2}{\sqrt{30}}, \quad \text{and} \quad B_z = \frac{1}{\sqrt{30}}$$

Therefore,

$$\underline{\underline{\vec{B} = \frac{1}{\sqrt{30}} (5\hat{a}_x - 2\hat{a}_y + \hat{a}_z)}}$$

(b) $\vec{B} \perp \vec{A}$, Angle between $\vec{A}$ & $\vec{B} = 90^\circ$

$$\vec{A} \cdot \vec{B} = 0$$

$$\vec{A} = 3\hat{a}_x - 2\hat{a}_y + \hat{a}_z, \quad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z. \quad (B_z=0) \quad (\vec{B} \text{ lies in the } xy \text{ plane})$$

$$\vec{A} \cdot \vec{B} = 3B_x - 2B_y = 0$$

combining with $B_x^2 + B_y^2 + B_z^2 = 1$

$$B_x = \frac{2}{\sqrt{29}}, \quad \text{and} \quad B_y = \frac{3}{\sqrt{29}}$$

therefore,

$$\underline{\underline{\vec{B} = \frac{1}{\sqrt{29}} (2\hat{a}_x + 3\hat{a}_y)}}$$